

Question number	Answer	Additional guidance	Mark
5(a)	Fick's (Law of Diffusion) (1)		(1)

Question number	Answer	Mark
5(b)	The only correct answer is C (it will halve) <i>A is incorrect because the value will halve</i> <i>B is incorrect because the value will halve</i> <i>D is incorrect because the value will halve</i>	(1)

Question number	Answer	Additional guidance	Mark
5(c)	A description that includes the following points: - lots of alveoli for large surface area (1) {single-celled / thin-walled} {alveoli / capillaries} for short diffusion distance (1) - good blood supply for high concentration gradient (1)	ACCEPT lots of capillaries for large surface area air sacs for alveoli ref to lots of alveoli <u>and</u> thin-walled <u>and</u> good blood supply = 1 mark if no other marks awarded	(3)

Question number	Answer
*5(d)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. Structure • salamanders have both lungs and their skin for <u>more</u> gas exchange • lungs allow oxygen from air and skin oxygen from water • the skin is folded to increase surface area • flattened body to increase surface area • so that gas exchange will be faster • skin has a good blood supply so that there will be high concentration gradient • skin is very permeable to gases so that they can diffuse in faster Behaviour • salamanders live in fast-flowing water which will have high levels of oxygen • shallow water will also have higher levels of oxygen • shallow water will make breathing with lungs easier • the rocking / swaying movements stir up the water • which helps to aerate the water • so that water with higher oxygen content is in contact with their skin • so that there is a higher concentration gradient • so that {gas exchange / diffusion of gases} will be faster • the rocking / swaying maintains the levels of oxygen in the blood Graph • at low levels of oxygen in the water the frequency of rocking is high • accompanied by an increase in blood pO_2 • as oxygen levels in the water increase the frequency of rocking decreases • but the blood pO_2 continues to increase • sufficient oxygen in water to diffuse into lungs and skin • to maintain levels of oxygen • rocking stops to conserve energy (6)

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.	Simple links made between gas exchange and information given or own knowledge 1 mark = 1 comment which can include a description of data 2 marks = 3 comments which can include a description of data
Level 2	3-4	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.	Extended links made between gas exchange and information given 3 marks = comments on both structure and behaviour with extended comment on one 4 marks = extended comments made on both structure and behaviour
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting and applying relevant biological facts / concepts. Consequences are discussed which supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.	Understanding of graph used to support discussion 5 marks = plus some discussion of graph to support answer 6 marks = plus an understanding of the relationships shown in the graph